

Exercise of Supervised Learning: Regularization Part 2

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Exercise 1

For a design matrix $\mathbf{X} \in \mathbb{R}^{n \times p}$ and the vector of targets $\mathbf{y} \in \mathbb{R}^n$, consider Lasso regression, i.e.,

$$\arg \min_{\boldsymbol{\theta} \in \mathbb{R}^p} 0.5 \|\mathbf{X}\boldsymbol{\theta} - \mathbf{y}\|_2^2 + \lambda \|\boldsymbol{\theta}\|_1.$$

where $\lambda > 0$ is the regularization parameter.

(a) Since there exists no analytical solution to Lasso regression in general, we want to find a procedure similar to gradient descent that should converge to the true solution.

Exercise 1 (a) (i)

Question (a) (i): Explain why $\mathcal{R}_{\text{reg}}$ is not differentiable.

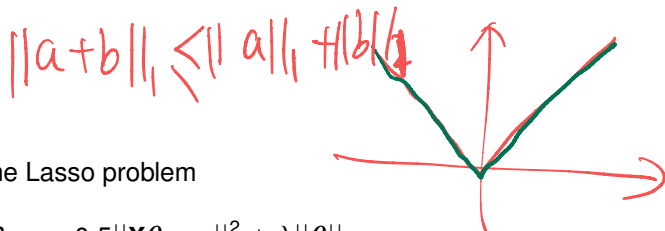
Solution: Because $\lambda \|\theta\|_1$ is not differentiable at $\theta = \mathbf{0}$.

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Exercise 1 (a) (ii)



Recall the regularized risk of the Lasso problem

$$\mathcal{R}_{\text{reg}} = 0.5 \|\mathbf{X}\boldsymbol{\theta} - \mathbf{y}\|_2^2 + \lambda \|\boldsymbol{\theta}\|_1.$$

Question (a) (ii): Show that $\mathcal{R}_{\text{reg}}$ is convex. *Hint: The sum of convex function is convex.*

Solution:

1. $0.5 \|\mathbf{X}\boldsymbol{\theta} - \mathbf{y}\|_2^2$ is convex, since it is quadratic.
2. $\lambda \|\boldsymbol{\theta}\|_1$ is also convex (since it is a norm).
3. Sum of convex functions is convex. So $\mathcal{R}_{\text{reg}}$ is convex.

$$\|\lambda u + (1-\lambda)v\|_1,$$



$$\lambda \|u\|_1 + (1-\lambda) \|v\|_1,$$

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$$0.5 \frac{\partial}{\partial \theta_j} \|\mathbf{x}\boldsymbol{\theta} - \mathbf{y}\|_2^2 = -\rho_j + \theta_j z_j.$$

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Solution:

$$0.5 \frac{\partial}{\partial \theta_j} \|\mathbf{x}\boldsymbol{\theta} - \mathbf{y}\|_2^2 = \frac{\partial}{\partial \theta_j} 0.5 \sum_{i=1}^n \left(y^{(i)} - \sum_{k=1}^p \mathbf{x}_k^{(i)} \theta_k \right)^2$$

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$$\frac{\partial \frac{1}{2} \sum_{i=1}^n r_i^2}{\partial \theta_j} = \frac{\partial r_i}{\partial \theta_j} \cdot r_i$$

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$$r_i = y^{(i)} - \sum_{k=1}^p x_k^{(i)} \theta_k$$

$$\left[\mathbf{X}^T (\mathbf{X}\boldsymbol{\theta} - \mathbf{y}) \right]_j$$

$p \times n$ $n \times 1$

$$x_j^{(i)}$$

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$$\sum_{k=1}^p x_k^{(i)} \theta_k$$

$$= x_1^{(i)} \theta_1 + x_2^{(i)} \theta_2 + \dots$$

$$+ \dots + x_j^{(i)} \theta_j + \dots$$

$$+ \dots + x_p^{(i)} \theta_p$$

Exercise 1 (a) (iii): matrix form, step by step

Question (a)(iii): find ρ_j, z_j with $0.5 \frac{\partial}{\partial \theta_j} \|\mathbf{X}\boldsymbol{\theta} - \mathbf{y}\|_2^2 = -\rho_j + \theta_j z_j$.

- ▶ *slope* $z_j = \|\mathbf{X}_{\cdot j}\|_2^2 \geq 0$: size (energy) of feature j .
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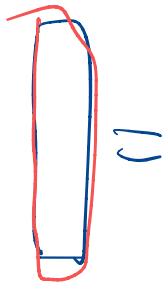
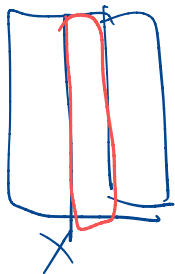
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$$- \underbrace{\mathbf{X}_{\cdot j}^\top (\mathbf{y} - \mathbf{X}_{-j}\boldsymbol{\theta}_{-j})}_{\rho_j}$$

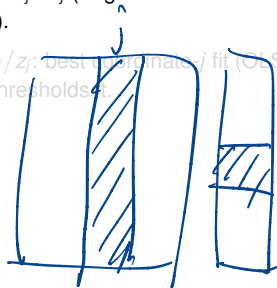
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$\rightarrow z_j$



$=$

$\mathbf{y} -$



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$$= -\rho_j + \theta_j z_j = 0 \Rightarrow \theta_j = \frac{\rho_j}{z_j}$$

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- ▶ *root* $\theta_j = \rho_j / z_j$: best coordinate- j fit (OLS); (a)(v) soft-thresholds it.

Exercise 1 (a): what we have so far

Goal: minimize $\mathcal{R}_{\text{reg}}(\boldsymbol{\theta}) = 0.5 \|\mathbf{X}\boldsymbol{\theta} - \mathbf{y}\|_2^2 + \lambda \|\boldsymbol{\theta}\|_1$, one coordinate θ_j at a time.

- ▶ (i) $\mathcal{R}_{\text{reg}}$ *not* differentiable — the kink of $\lambda \|\boldsymbol{\theta}\|_1$ at $\boldsymbol{\theta} = \mathbf{0}$.
- ▶ (ii) but $\mathcal{R}_{\text{reg}}$ *convex* — a per-coordinate minimum is global.
- ▶ (iii) the smooth part is linear in θ_j : $0.5 \frac{\partial}{\partial \theta_j} \|\mathbf{X}\boldsymbol{\theta} - \mathbf{y}\|_2^2 = -\rho_j + \theta_j z_j$,
with slope $z_j = \|\mathbf{X}_{\cdot j}\|_2^2$ and $\rho_j = \mathbf{X}_{\cdot j}^\top \mathbf{r}_{-j}$ (partial residual).

Next (a)(iv): only $\lambda |\theta_j|$ remains, non-smooth at 0. Use its *subderivative* and add to (iii) to assemble $\partial_{\theta_j} \mathcal{R}_{\text{reg}}$.

$$\lambda \|\boldsymbol{\theta}\|_1 = \lambda \|\theta_1\|_1 + \lambda \|\theta_2\|_1 + \dots + \lambda \|\theta_j\|_1 + \dots$$
$$\lambda \|\theta_j\|_1$$

Exercise 1 (a) (iv)

Question (a) (iv): In this situation, we can use the so-called sub-derivative which we denote with ∂f for a real-valued convex continuous function f . The subderivative maps a point $\theta \in \mathbb{R}$ to an interval

► and if f is differentiable at $\tilde{\theta} \in \mathbb{R}$, then $\partial f(\tilde{\theta}) = \left\{ \frac{d}{d\theta} f(\tilde{\theta}) \right\}$,

► and for $f(\theta) = \lambda|\theta|$ and $\lambda > 0$, it holds that $\partial f(\theta) = \begin{cases} \{-\lambda\} & \text{for } \theta < 0, \\ [-\lambda, \lambda] & \text{for } \theta = 0, \\ \{\lambda\} & \text{for } \theta > 0 \end{cases}$

► and for f, g real-valued convex functions with $\partial f(\tilde{\theta}) = [a, b]$, $\partial g(\tilde{\theta}) = [c, d]$,

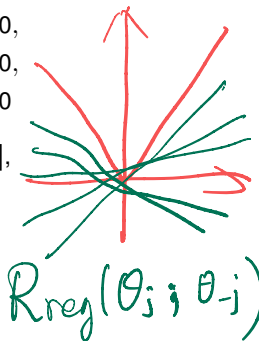
$$\partial(f + g)(\tilde{\theta}) = [a + c, b + d]$$

where $b \geq a$ and $d \geq c$.

With this compute the sub-derivative of $\mathcal{R}_{\text{reg}, \theta_{\neq j}}$ w.r.t. θ_j , i.e., $\partial_{\theta_j} \mathcal{R}_{\text{reg}, \theta_{\neq j}}$ where

$\mathcal{R}_{\text{reg}, \theta_{\neq j}} : \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}, \theta_j \mapsto \mathcal{R}_{\text{reg}}(\theta_1, \dots, \theta_j, \dots, \theta_p)$ for constants

$\theta_{\neq j} = (\theta_1, \dots, \theta_{j-1}, \theta_{j+1}, \dots, \theta_p)^T$. Hint: Use (a) (iii).



Fixed!

Exercise 1 (a) (iv): case by case

 Σ

Solution: by the sum rule $\partial_{\theta_j} \mathcal{R}_{\text{reg}, \theta_{\neq j}} = \underbrace{\{-\rho_j + \theta_j z_j\}}_{\text{from (iii)}} + \partial(\lambda|\theta_j|)$, and $\partial(\lambda|\theta_j|)$ splits on the

sign of θ_j :

θ_j	smooth part	$\partial(\lambda \theta_j)$	$\partial_{\theta_j} \mathcal{R}_{\text{reg}}$
< 0	$\{-\rho_j + \theta_j z_j\}$	$\{-\lambda\}$	
$= 0$	$\{-\rho_j\}$	$[-\lambda, \lambda]$	
> 0	$\{-\rho_j + \theta_j z_j\}$	$\{\lambda\}$	

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$= 0$	$\{-\rho_j\}$	$[-\lambda, \lambda]$	
> 0	$\{-\rho_j + \theta_j z_j\}$	$\{\lambda\}$	

Exercise 1 (a) (iv): case by case

Solution: by the sum rule $\partial_{\theta_j} \mathcal{R}_{\text{reg}, \theta_{\neq j}} = \underbrace{\{-\rho_j + \theta_j z_j\}}_{\text{from (iii)}} + \partial(\lambda|\theta_j|)$, and $\partial(\lambda|\theta_j|)$ splits on the

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$= 0$	$\{-\rho_j\}$	$[-\lambda, \lambda]$	$[-\rho_j - \lambda, -\rho_j + \lambda]$
> 0	$\{-\rho_j + \theta_j z_j\}$	$\{\lambda\}$	

Exercise 1 (a) (iv): case by case

Solution: by the sum rule $\partial_{\theta_j} \mathcal{R}_{\text{reg}, \theta_{\neq j}} = \underbrace{\{-\rho_j + \theta_j z_j\}}_{\text{from (iii)}} + \partial(\lambda|\theta_j|)$, and $\partial(\lambda|\theta_j|)$ splits on the

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Solution: by the sum rule $\partial_{\theta_j} \mathcal{R}_{\text{reg}, \theta_{\neq j}} = \underbrace{\{-\rho_j + \theta_j z_j\}}_{\text{from (iii)}} + \partial(\lambda|\theta_j|)$, and $\partial(\lambda|\theta_j|)$ splits on the

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> 0	$\{-\rho_j + \theta_j z_j\}$	$\{\lambda\}$	$\{-\rho_j + \theta_j z_j + \lambda\}$

The right column is $\partial_{\theta_j} \mathcal{R}_{\text{reg}, \theta_{\neq j}}$ — the optimality input for (a)(v).

Exercise 1 (a) (v)

Question (a) (v): For a real-valued convex function f , the global minimum (if it exists) can be characterized in the following way:

A point $\theta^* \in \mathbb{R}$ is the global minimum of f if and only if $0 \in \partial f(\theta^*)$.

With this show that

$$\theta_j^* \in \arg \min_{\theta_j \in \mathbb{R}} \mathcal{R}_{\text{reg}, \theta_{\neq j}} \Leftrightarrow \theta_j^* = \begin{cases} \frac{\rho_j + \lambda}{z_j} & \text{for } \rho_j < -\lambda \\ 0 & \text{for } -\lambda \leq \rho_j \leq \lambda \\ \frac{\rho_j - \lambda}{z_j} & \text{for } \rho_j > \lambda \end{cases}$$

Exercise 1 (a) (v): Solve $0 \in \partial_{\theta_j} \mathcal{R}_{\text{reg}}$ in each sign-case

Solution: set $0 \in \partial_{\theta_j} \mathcal{R}_{\text{reg}, \theta_{\neq j}}$, the subgradient from (a)(iv) (with $z_j = \|\mathbf{X}_{\cdot j}\|_2^2 > 0$):

$$\partial_{\theta_j} \mathcal{R}_{\text{reg}, \theta_{\neq j}} = \begin{cases} \{-\rho_j + \theta_j z_j - \lambda\} & \theta_j < 0, \\ [-\rho_j - \lambda, -\rho_j + \lambda] & \theta_j = 0, \\ \{-\rho_j + \theta_j z_j + \lambda\} & \theta_j > 0. \end{cases}$$

Solve each case for θ_j^* . For the singletons, $0 \in \{x\}$ means $0 = x$; the assumed sign then forces a condition on ρ_j :

- ▶ $\theta_j < 0$: $0 = -\rho_j + \theta_j z_j - \lambda \Rightarrow \theta_j^* = \frac{\rho_j + \lambda}{z_j}$; with $z_j > 0$, $\theta_j^* < 0 \Rightarrow \rho_j < -\lambda$.
- ▶ $\theta_j = 0$: $0 \in [-\rho_j - \lambda, -\rho_j + \lambda] \Rightarrow -\rho_j - \lambda \leq 0 \leq -\rho_j + \lambda \Rightarrow -\lambda \leq \rho_j \leq \lambda$.
- ▶ $\theta_j > 0$: $0 = -\rho_j + \theta_j z_j + \lambda \Rightarrow \theta_j^* = \frac{\rho_j - \lambda}{z_j}$; with $z_j > 0$, $\theta_j^* > 0 \Rightarrow \rho_j > \lambda$.

Exercise 1 (a) (v): Solve $0 \in \partial_{\theta_j} \mathcal{R}_{\text{reg}}$ in each sign-case

Solution: set $0 \in \partial_{\theta_j} \mathcal{R}_{\text{reg}, \theta_{\neq j}}$, the subgradient from (a)(iv) (with $z_j = \|\mathbf{X}_{\cdot j}\|_2^2 > 0$):

$$\partial_{\theta_j} \mathcal{R}_{\text{reg}, \theta_{\neq j}} = \begin{cases} \{-\rho_j + \theta_j z_j - \lambda\} & \theta_j < 0, \\ [-\rho_j - \lambda, -\rho_j + \lambda] & \theta_j = 0, \\ \{-\rho_j + \theta_j z_j + \lambda\} & \theta_j > 0. \end{cases}$$

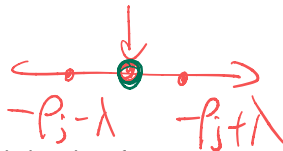
Solve each case for θ_j^* . For the singletons, $0 \in \{x\}$ means $0 = x$; the assumed sign then forces a condition on ρ_j :

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Exercise 1 (a) (v): Solve $0 \in \partial_{\theta_j} \mathcal{R}_{\text{reg}}$ in each sign-case

Solution: set $0 \in \partial_{\theta_j} \mathcal{R}_{\text{reg}, \theta_{\neq j}}$, the subgradient from (a)(iv) (with $z_j = \|\mathbf{X}_{\cdot j}\|_2^2 > 0$):

$$\partial_{\theta_j} \mathcal{R}_{\text{reg}, \theta_{\neq j}} = \begin{cases} \{-\rho_j + \theta_j z_j - \lambda\} & \theta_j < 0, \\ [-\rho_j - \lambda, -\rho_j + \lambda] & \theta_j = 0, \\ \{-\rho_j + \theta_j z_j + \lambda\} & \theta_j > 0. \end{cases}$$



Solve each case for θ_j^* . For the singletons, $0 \in \{x\}$ means $0 = x$; the assumed sign then forces a condition on ρ_j :

- $\theta_j < 0$: $0 = -\rho_j + \theta_j z_j - \lambda \Rightarrow \theta_j^* = \frac{\rho_j + \lambda}{z_j}$; with $z_j > 0$, $\theta_j^* < 0 \Rightarrow \rho_j < -\lambda$.
- $\theta_j = 0$: $0 \in [-\rho_j - \lambda, -\rho_j + \lambda] \Rightarrow -\rho_j - \lambda \leq 0 \leq -\rho_j + \lambda \Rightarrow -\lambda \leq \rho_j \leq \lambda$.
- $\theta_j > 0$: $0 = -\rho_j + \theta_j z_j + \lambda \Rightarrow \theta_j^* = \frac{\rho_j - \lambda}{z_j}$; with $z_j > 0$, $\theta_j^* > 0 \Rightarrow \rho_j > \lambda$.

Exercise 1 (a) (v): Solve $0 \in \partial_{\theta_j} \mathcal{R}_{\text{reg}}$ in each sign-case

Solution: set $0 \in \partial_{\theta_j} \mathcal{R}_{\text{reg}, \theta_{\neq j}}$, the subgradient from (a)(iv) (with $z_j = \|\mathbf{X}_{\cdot j}\|_2^2 > 0$):

$$\partial_{\theta_j} \mathcal{R}_{\text{reg}, \theta_{\neq j}} = \begin{cases} \{-\rho_j + \theta_j z_j - \lambda\} & \theta_j < 0, \\ [-\rho_j - \lambda, -\rho_j + \lambda] & \theta_j = 0, \\ \{-\rho_j + \theta_j z_j + \lambda\} & \theta_j > 0. \end{cases}$$

Handwritten notes:

$$\frac{\rho_j + \lambda}{z_j} < 0$$
$$\Downarrow$$
$$\rho_j + \lambda < 0$$
$$\rho_j < -\lambda$$

Solve each case for θ_j^* . For the singletons, $0 \in \{x\}$ means $0 = x$; the assumed sign then forces a condition on ρ_j :

- $\theta_j < 0$: $0 = -\rho_j + \theta_j z_j - \lambda \Rightarrow \theta_j^* = \frac{\rho_j + \lambda}{z_j}$; with $z_j > 0$, $\theta_j^* < 0 \Rightarrow \rho_j < -\lambda$.
- $\theta_j = 0$: $0 \in [-\rho_j - \lambda, -\rho_j + \lambda] \Rightarrow -\rho_j - \lambda \leq 0 \leq -\rho_j + \lambda \Rightarrow -\lambda \leq \rho_j \leq \lambda$.
- $\theta_j > 0$: $0 = -\rho_j + \theta_j z_j + \lambda \Rightarrow \theta_j^* = \frac{\rho_j - \lambda}{z_j}$; with $z_j > 0$, $\theta_j^* > 0 \Rightarrow \rho_j > \lambda$.

Exercise 1 (a) (v): Continued

Summarize everything up:

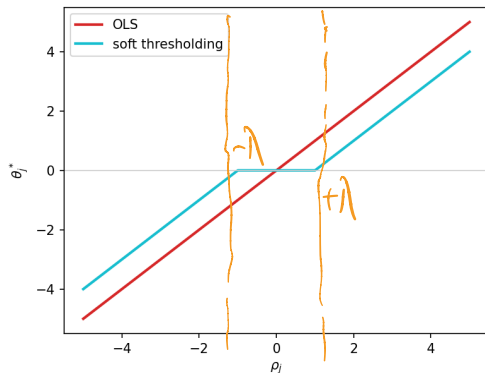
$$\theta_j^* \in \arg \min_{\theta_j \in \mathbb{R}} \mathcal{R}_{\text{reg}, \theta_{\neq j}} \Leftrightarrow \theta_j^* = \begin{cases} \frac{\rho_j + \lambda}{z_j} & \text{for } \rho_j < -\lambda \\ 0 & \text{for } -\lambda \leq \rho_j \leq \lambda \\ \frac{\rho_j - \lambda}{z_j} & \text{for } \rho_j > \lambda \end{cases}$$

Exercise 1 (a) (vi)

Question (a) (vi): Plot θ_j^* as a function of ρ_j for $\rho_j \in [-5, 5]$, $\lambda = 1$, $z_j = 1$. (This function is called the soft thresholding operator.)

```
1 rhos = np.arange(-5, 5.001, 0.1)
2 lam = z = 1.0
3 theta_star = np.where(
4     rhos < -lam, (rhos + lam) / z,
5     np.where(rhos > lam,
6             (rhos - lam) / z, 0.0))
7 theta_ols = rhos / z
```

Soft thresholding is OLS pulled toward 0 by λ , and clamped to exactly 0 on $[-\lambda, \lambda]$ — the source of sparsity.



Exercise 1 (b)

Question (b): for non-singular $\mathbf{X}^\top \mathbf{X}$, find $\mathbf{A} \in \mathbb{R}^{p \times p}$ with

$$\mathbf{A}^\top \mathbf{X}^\top \mathbf{X} \mathbf{A} = \mathbf{I}.$$

Hint: $\mathbf{X}^\top \mathbf{X}$ is positive definite.

Use the hint: what structure does positive-definiteness give us?

Recall — a positive definite matrix has a spectral decomposition

$$\mathbf{X}^\top \mathbf{X} = \mathbf{V} \mathbf{D} \mathbf{V}^\top, \quad \mathbf{V}^\top \mathbf{V} = \mathbf{I}, \quad \mathbf{D} = \text{diag}(d_1, \dots, d_p), \quad d_i > 0.$$

Exercise 1 (b)

Question (b): for non-singular $\mathbf{X}^\top \mathbf{X}$, find $\mathbf{A} \in \mathbb{R}^{p \times p}$ with

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Exercise 1 (b): solve for $\mathbf{A}$

Want $\mathbf{A}^\top \mathbf{X}^\top \mathbf{X} \mathbf{A} = \mathbf{I}$, using $\mathbf{X}^\top \mathbf{X} = \mathbf{V} \mathbf{D} \mathbf{V}^\top$.

- ▶ plug in: $\mathbf{A}^\top \mathbf{V} \mathbf{D} \mathbf{V}^\top \mathbf{A} = \mathbf{I}$
- ▶ substitute $\mathbf{P} := \mathbf{V}^\top \mathbf{A}$: $\mathbf{P}^\top \mathbf{D} \mathbf{P} = \mathbf{I}$
- ▶ $\mathbf{D}$ diagonal, $d_i > 0 \Rightarrow \mathbf{P} = \mathbf{D}^{-0.5}$ (then $\mathbf{D}^{-0.5} \mathbf{D} \mathbf{D}^{-0.5} = \mathbf{I}$)
- ▶ undo the substitution: $\mathbf{V}^\top \mathbf{A} = \mathbf{D}^{-0.5}$, $\mathbf{V}$ orthogonal $\Rightarrow \mathbf{A} = \mathbf{V} \mathbf{D}^{-0.5}$

Exercise 1 (b): solve for $\mathbf{A}$

$$\underbrace{\mathbf{A}^T \mathbf{V} \mathbf{D}}_{\mathbf{P}^T} \underbrace{\mathbf{V}^T \mathbf{A}}_{\mathbf{P}} = \mathbf{I}$$

Want $\mathbf{A}^T \mathbf{X}^T \mathbf{X} \mathbf{A} = \mathbf{I}$, using $\mathbf{X}^T \mathbf{X} = \mathbf{V} \mathbf{D} \mathbf{V}^T$.

- ▶ plug in: $\mathbf{A}^T \mathbf{V} \mathbf{D} \mathbf{V}^T \mathbf{A} = \mathbf{I}$
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- ▶ undo the substitution: $\mathbf{V}^T \mathbf{A} = \mathbf{D}^{-0.5}$, $\mathbf{V}$ orthogonal $\Rightarrow \mathbf{A} = \mathbf{V} \mathbf{D}^{-0.5}$

Exercise 1 (b): solve for $\mathbf{A}$

Want $\mathbf{A}^\top \mathbf{X}^\top \mathbf{X} \mathbf{A} = \mathbf{I}$, using $\mathbf{X}^\top \mathbf{X} = \mathbf{V} \mathbf{D} \mathbf{V}^\top$.

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- ▶ undo the substitution: $\mathbf{V}^\top \mathbf{A} = \mathbf{D}^{-0.5}$, $\mathbf{V}$ orthogonal $\Rightarrow \mathbf{A} = \mathbf{V} \mathbf{D}^{-0.5}$

Exercise 1 (b): solve for A

Want $\mathbf{A}^\top \mathbf{X}^\top \mathbf{X} \mathbf{A} = \mathbf{I}$, using $\mathbf{X}^\top \mathbf{X} = \mathbf{V} \mathbf{D} \mathbf{V}^\top$.

- ▶ plug in: $\mathbf{A}^\top \mathbf{V} \mathbf{D} \mathbf{V}^\top \mathbf{A} = \mathbf{I}$
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- ▶ $\mathbf{D}$ diagonal, $d_i > 0 \Rightarrow \mathbf{P} = \mathbf{D}^{-0.5}$ (then $\mathbf{D}^{-0.5} \mathbf{D} \mathbf{D}^{-0.5} = \mathbf{I}$)
- ▶ undo the substitution: $\mathbf{V}^\top \mathbf{A} = \mathbf{D}^{-0.5}$, $\mathbf{V}$ orthogonal $\Rightarrow \mathbf{A} = \mathbf{V} \mathbf{D}^{-0.5}$

$$\mathbf{V} \mathbf{P} = \mathbf{V} \mathbf{V}^\top \mathbf{A}$$

$$\mathbf{A} = \mathbf{V} \mathbf{P} = \mathbf{V} \cdot \mathbf{D}^{-0.5}$$

Exercise 1 (b): solve for $\mathbf{A}$

Want $\mathbf{A}^\top \mathbf{X}^\top \mathbf{X} \mathbf{A} = \mathbf{I}$, using $\mathbf{X}^\top \mathbf{X} = \mathbf{V} \mathbf{D} \mathbf{V}^\top$.

- ▶ plug in: $\mathbf{A}^\top \mathbf{V} \mathbf{D} \mathbf{V}^\top \mathbf{A} = \mathbf{I}$
- ▶ substitute $\mathbf{P} := \mathbf{V}^\top \mathbf{A}$: $\mathbf{P}^\top \mathbf{D} \mathbf{P} = \mathbf{I}$
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- ▶ undo the substitution: $\mathbf{V}^\top \mathbf{A} = \mathbf{D}^{-0.5}$, $\mathbf{V}$ orthogonal $\Rightarrow \mathbf{A} = \mathbf{V} \mathbf{D}^{-0.5}$

V is orthonormal

Exercise 1 (c) (i)

(c) For a design matrix with orthonormal columns, i.e., $\mathbf{X}^T \mathbf{X} = \mathbf{I}$, exists an analytical minimizer of the Lasso regression $\hat{\boldsymbol{\theta}}_{\text{Lasso}} = (\hat{\theta}_{\text{Lasso},1}, \dots, \hat{\theta}_{\text{Lasso},p})^T$ that is given by

$$\hat{\theta}_{\text{Lasso},i} = \text{sign}(\hat{\theta}_i) \max \left\{ \left| \hat{\theta}_i \right| - \lambda, 0 \right\}, \quad i = 1, \dots, p,$$

where $\hat{\boldsymbol{\theta}} = (\hat{\theta}_1, \dots, \hat{\theta}_p)^T = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$ is the minimizer of the unregularized empirical risk (w.r.t. the L2 loss).

Under the assumption that $\mathbf{X}^T \mathbf{X}$ is non-singular, your colleague proposes to project $\mathbf{X}$ with $\mathbf{A}$ from (b), i.e., use $\tilde{\mathbf{X}} = \mathbf{X}\mathbf{A}$ and then apply the analytical solution given here.

(i) Show that this does not generally solve the original Lasso regression.

Hint: You only need to check under which condition

$\nabla_{\boldsymbol{\theta}} 0.5 \|\mathbf{X}\boldsymbol{\theta} - \mathbf{y}\|_2^2 = \nabla_{\boldsymbol{\theta}} 0.5 \|\mathbf{X}\mathbf{A}\boldsymbol{\theta} - \mathbf{y}\|_2^2$. The proof can be finished with a subgradient argument regarding stationarity, which you do not have to do.

Exercise 1 (c) (i): Continued

Solution: compare the two gradients (the hint).

$$\nabla_{\theta} 0.5 \|\mathbf{XA}\theta - \mathbf{y}\|_2^2 = \mathbf{A}^T \mathbf{X}^T \mathbf{XA} \theta - \mathbf{A}^T \mathbf{X}^T \mathbf{y}$$

$$\nabla_{\theta} 0.5 \|\mathbf{X}\theta - \mathbf{y}\|_2^2 = \mathbf{X}^T \mathbf{X} \theta - \mathbf{X}^T \mathbf{y}$$

equal $\Rightarrow \mathbf{A}^T \mathbf{X}^T \mathbf{XA} = \mathbf{X}^T \mathbf{X} \xrightarrow{(b)} \mathbf{X}^T \mathbf{X} = \mathbf{I}$, i.e. $\mathbf{A} = \mathbf{I}$. $\mathbf{A} \neq \mathbf{I} \Rightarrow$ projected solution $\neq$ original Lasso.

Exercise 1 (c) (i): Continued

$$\begin{aligned}\nabla_{\theta} \|X\theta - y\|_2^2 \\ = X^T(X\theta - y)\end{aligned}$$

Solution: compare the two gradients (the hint).

$$\begin{aligned}\nabla_{\theta} 0.5\|XA\theta - y\|_2^2 &= A^T \underline{X^T X A} \theta - A^T X^T y \\ \nabla_{\theta} 0.5\|X\theta - y\|_2^2 &= X^T X \theta - X^T y\end{aligned}$$

equal $\Rightarrow A^T X^T X A = X^T X \xrightarrow{(b)} X^T X = I$, i.e. $A = I$. $A \neq I \Rightarrow$ projected solution $\neq$ original Lasso.

$$\begin{aligned}\nabla_{\theta} \|\tilde{X}\theta - y\|_2^2 &= \tilde{X}^T (\tilde{X}\theta - y) \\ &= (XA)^T (XA\theta - y) \\ &= A^T X^T X A \theta - A^T X^T y\end{aligned}$$

Exercise 1 (c) (i): Continued

Solution: compare the two gradients (the hint).

$$\nabla_{\theta} 0.5 \|\mathbf{XA}\theta - \mathbf{y}\|_2^2 = \mathbf{A}^\top \mathbf{X}^\top \mathbf{XA} \theta - \mathbf{A}^\top \mathbf{X}^\top \mathbf{y}$$

$$\nabla_{\theta} 0.5 \|\mathbf{X}\theta - \mathbf{y}\|_2^2 = \mathbf{X}^\top \mathbf{X} \theta - \mathbf{X}^\top \mathbf{y}$$

equal $\Rightarrow \mathbf{A}^\top \mathbf{X}^\top \mathbf{XA} = \mathbf{X}^\top \mathbf{X} \xrightarrow{(b)} \mathbf{X}^\top \mathbf{X} = \mathbf{I}$, i.e. $\mathbf{A} = \mathbf{I}$. $\mathbf{A} \neq \mathbf{I} \Rightarrow$ projected solution $\neq$ original Lasso.

Exercise 1 (c) (i): Continued

Solution: compare the two gradients (the hint).

$$\nabla_{\theta} 0.5 \|\mathbf{XA}\theta - \mathbf{y}\|_2^2 = \mathbf{A}^T \mathbf{X}^T \mathbf{XA} \theta - \mathbf{A}^T \mathbf{X}^T \mathbf{y}$$

$$\nabla_{\theta} 0.5 \|\mathbf{X}\theta - \mathbf{y}\|_2^2 = \mathbf{X}^T \mathbf{X} \theta - \mathbf{X}^T \mathbf{y}$$

equal $\Rightarrow \mathbf{A}^T \mathbf{X}^T \mathbf{XA} = \mathbf{X}^T \mathbf{X} \xrightarrow{(b)} \mathbf{X}^T \mathbf{X} = \mathbf{I}$, i.e. $\mathbf{A} = \mathbf{I}$. $\mathbf{A} \neq \mathbf{I} \Rightarrow$ projected solution $\neq$ original Lasso.

Exercise 1 (c) (i): Continued

Solution: compare the two gradients (the hint).

$$\nabla_{\theta} 0.5 \|\mathbf{XA}\theta - \mathbf{y}\|_2^2 = \mathbf{A}^T \mathbf{X}^T \mathbf{XA} \theta - \mathbf{A}^T \mathbf{X}^T \mathbf{y}$$

$$\nabla_{\theta} 0.5 \|\mathbf{X}\theta - \mathbf{y}\|_2^2 = \mathbf{X}^T \mathbf{X} \theta - \mathbf{X}^T \mathbf{y}$$

equal $\Rightarrow \mathbf{A}^T \mathbf{X}^T \mathbf{XA} = \mathbf{X}^T \mathbf{X} \xrightarrow{(b)} \mathbf{X}^T \mathbf{X} = \mathbf{I}$, i.e. $\mathbf{A} = \mathbf{I}$. $\mathbf{A} \neq \mathbf{I} \Rightarrow$ projected solution $\neq$ original Lasso.

Exercise 1 (c) (ii)

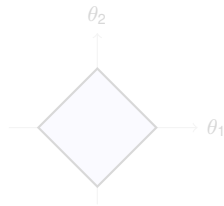
$$\|X\theta - y\|_2^2 + \lambda \|\theta\|_1$$

$$\|XA\theta - y\|_2^2 + \lambda \|\theta\|_1$$

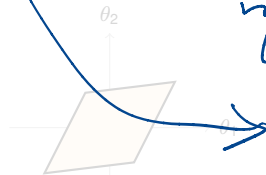
Question (c) (ii): How to adapt the penalty so the projected problem matches the original Lasso? Can we still solve coordinate-wise?

Solution:

- ▶ penalize $\|A\theta\|_1$ (not $\|\theta\|_1$) $\Rightarrow \theta_{\text{proj}}^* = A^{-1}\theta_{\text{Lasso}}^*$ — same minimal risk, equivalent.
- ▶ but $\|A\theta\|_1$ couples the coordinates $\Rightarrow$ no per-coordinate split — not solvable independently.



$\|\theta\|_1 \leq t$: axis-aligned
separable, sparse corners



$\|A\theta\|_1 \leq t$: tilted
coupled, corners off-axis

$$\|XA\tilde{\theta} - y\|_2^2 + \lambda \|A\tilde{\theta}\|_1$$

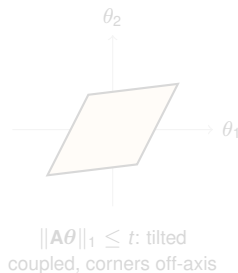
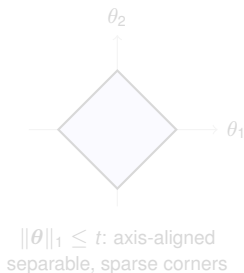
$$\begin{aligned} \tilde{\theta}_{\text{Lasso}}^* &= A\theta_{\text{proj}}^* \\ \theta_{\text{proj}}^* &= A^{-1}\tilde{\theta}_{\text{Lasso}}^* \end{aligned}$$

Exercise 1 (c) (ii)

Question (c) (ii): How to adapt the penalty so the projected problem matches the original Lasso? Can we still solve coordinate-wise?

Solution:

- ▶ penalize $\|\mathbf{A}\boldsymbol{\theta}\|_1$ (not $\|\boldsymbol{\theta}\|_1$) $\Rightarrow \boldsymbol{\theta}_{\text{proj}}^* = \mathbf{A}^{-1}\boldsymbol{\theta}_{\text{Lasso}}$ — same minimal risk, equivalent.
- ▶ but $\|\mathbf{A}\boldsymbol{\theta}\|_1$ couples the coordinates $\Rightarrow$ no per-coordinate split — not solvable independently.

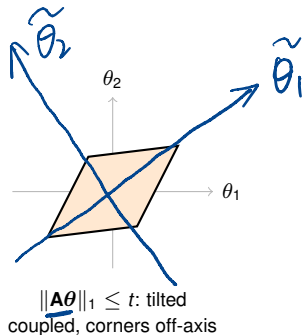
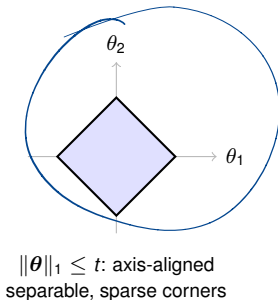


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Exercise 1 (c) (iii)

Question (c) (iii): Does the procedure in (c) perform variable selection?

Solution:

- ▶ $||\tilde{\mathbf{X}}\boldsymbol{\theta} - \mathbf{y}||_2^2 + \lambda||\boldsymbol{\theta}||_1$ with $\tilde{\mathbf{X}} = \mathbf{XA}$ — Lasso in the *projected* coordinates.
- ▶ so $\boldsymbol{\theta}^*$ is sparse on $\tilde{\mathbf{X}}$ — selects *projected* features.
- ▶ but $\mathbf{XA}\boldsymbol{\theta}^* = \mathbf{X}(\mathbf{A}\boldsymbol{\theta}^*)$, and $\mathbf{A}\boldsymbol{\theta}^*$ is generally dense $\Rightarrow$ no selection on $\mathbf{X}$.



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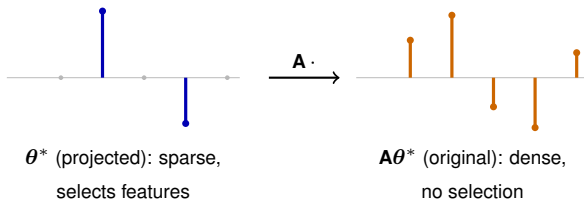
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$$\underbrace{\mathbf{XA}}_{\tilde{\mathbf{X}}} \boldsymbol{\theta}$$



Exercise 1 (d)

(d) You are given the following code to compare the quality of the projected Lasso regression vs. the regular Lasso regression. Complete the missing code of the algorithms and interpret the result.

The two estimators are ported from the R reference solution to Python. Part (b) gives $\mathbf{A} = \mathbf{V}\mathbf{D}^{-0.5}$, which orthonormalizes the design so the closed-form Lasso applies; the regular Lasso uses the coordinate-descent update derived in (a)(v).

Exercise 1 (d): the two estimators (Python)

```
1 def proj_orth_lasso(X, y, lam):
2     vals, vecs = np.linalg.eigh(X.T @ X)
3     A = vecs @ np.diag(vals ** -0.5)
4     Xt = X @ A           #  $Xt'Xt = I$ 
5     b = Xt.T @ y         # OLS, orthonormal
6     b = np.sign(b) * np.maximum(
7         np.abs(b) - lam, 0.0) # soft-threshold
8     return A @ b         # back to X-coords
```

```
1 def lasso(X, y, lam, N):
2     p = X.shape[1]
3     theta = np.ones(p)
4     for i in range(1, N + 1):
5         j = i % p
6         mask = np.ones(p); mask[j] = 0.0
7         rho = X[:, j] @ (y - X @ (theta * mask))
8         z = np.sum(X[:, j] ** 2)
9         if rho < -lam: theta[j] = (rho+lam)/z
10        elif rho > lam: theta[j] = (rho-lam)/z
11        else:         theta[j] = 0.0
12    return theta
```

Left: project $\rightarrow$ closed-form soft-threshold $\rightarrow$ map back via **A**. Right: cyclic coordinate descent applying the $(a)(v)$ update to one θ_j at a time.

Exercise 1 (d): the experiment

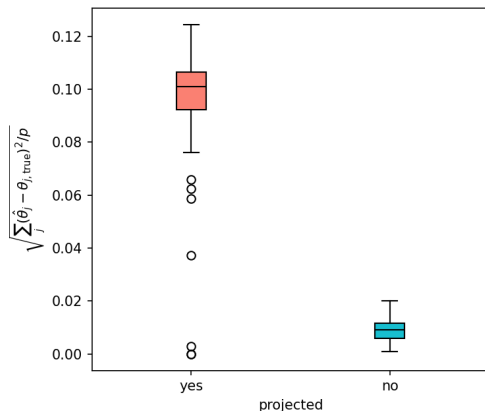
```
1 p, n, N = 10, 100, 400
2 lam, s_sig, s_noise = 1.0, 1.0, 0.1
3 rmse = lambda e, t: n/(n-1)*np.std(e-t, ddof=1)
4 for _ in range(100):
5     X = rng.normal(scale=s_sig, size=(n, p))
6     th = rng.normal(size=p)
7     th[rng.binomial(1, 0.7, p) == 1] = 0 # sparse
8     y = X @ th + rng.normal(scale=s_noise, size=n)
9     # compare rmse(proj_orth_lasso) vs rmse(lasso)
```

Setup (100 repetitions):

- ▶ random design $\mathbf{X}$ ($n=100$, $p=10$)
- ▶ true θ : $\sim 70\%$ of entries *forced* to 0 — sparse in the original coordinates
- ▶ $\mathbf{y} = \mathbf{X}\theta + \text{small noise}$

Metric: RMSE = how far the estimate $\hat{\theta}$ is from the true θ (lower = better recovery). **Compare:** proj_orth_lasso (projected) vs. lasso (regular).

Exercise 1 (d): result



“yes” = projected, “no” = regular

Regular (≈ 0.01) beats projected (≈ 0.10) by $\sim 10\times$.

Why?

- ▶ regular penalizes in the original coords $\Rightarrow$ zeros the junk $\Rightarrow$ recovers the sparse θ .
- ▶ projected is sparse in rotated coords; back in original, $\hat{\theta} = \mathbf{A} \cdot (\text{sparse})$ is *dense* $\Rightarrow$ misses the sparse θ .

Confirms (c): the projection selects in the *wrong* coordinate system.

Summary — Connecting the exercise to the lecture

	What we derived	Lecture concept made concrete
(a)	$\mathcal{R}_{\text{reg}}$ convex but non-smooth; coordinate-wise minimizer $\theta_j^* = \text{ST}_\lambda(\rho_j)/z_j$ (soft-thresholding)	<i>Soft-thresholding and L1 regularization</i> ; coordinate descent for Lasso
(b)	$\mathbf{A} = \mathbf{V}\mathbf{D}^{-0.5}$ orthonormalizes the design: $\mathbf{A}^\top \mathbf{X}^\top \mathbf{X} \mathbf{A} = \mathbf{I}$	<i>L1 and L2 Reg. with orthonormal design</i> (where Lasso has a closed form)
(c)	projected closed form $\neq$ original Lasso; it selects in rotated, not original, coordinates	<i>L1 geometry</i> : sparsity is <i>basis-dependent</i>
(d)	regular Lasso recovers the sparse θ ; projected $\sim 10\times$ worse	<i>Support recovery of Lasso</i> / feature selection

Take-aways

- ▶ no general closed form $\Rightarrow$ coordinate descent; the per-coordinate step is *soft-thresholding*.
- ▶ orthonormal design ($\mathbf{X}^\top \mathbf{X} = \mathbf{I}$) is the special case with an exact soft-threshold formula.
- ▶ L1 selects only in the basis it penalizes; rotating the design leaves $\hat{\theta}$ dense in the original coordinates.

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